

Project: Soluble Unit Dose (SUD) Rejects and Equipment Performance Analysis

Project Description:

During my time at DataStories, I collaborated with Procter & Gamble to process large-scale datasets related to Soluble Unit Dose (SUD) rejects and equipment performance across multiple production lines. The primary goal was to provide actionable insights that optimize manufacturing processes and reduce reject rates.

This script demonstrates the implementation of an ETL pipeline for the SUB Box Tool, one of the tools I developed at Procter & Gamble (P&G). This script involves getting data from p&g server, transforming data and creating fact and dimension tables for data model in Power bi and developing interactive dashboard.

Key Contributions:

- Developed and implemented ETL pipelines to process and clean large volumes of production data, ensuring accuracy and efficiency in data analytics.
- Designed and developed data models and interactive Power BI dashboards to visualize trends in reject rates, equipment performance, and operational inefficiencies.
- Collaborated with cross-functional teams to optimize data models and workflows, ensuring that the insights provided aligned with business requirements.
- Identified recurring patterns in reject rates and equipment malfunctions, enabling resolution of issues that led to manufacturing inefficiencies.
- Analysed SUD reject rates and operational trends across multiple sites and production lines for deeper insights.

The Power BI dashboard I developed for P&G was deployed within their enterprise Power BI environment, which included integration with various data sources from P&G's internal systems, such as production databases and real-time equipment performance data.

In this script I implemented incremental data loading after initial data loading ensured that we didn't overburden the system with unnecessary data, making the ETL pipeline more efficient. The pipeline was automated using Task Scheduler and configured Power BI for scheduled refreshes, ensuring stakeholders always had access to the latest data insights.

Few examples of metrics which I reported on power BI dashboard are as below:

- Reject Percentage: The percentage of Soluble Unit Dose (SUD) units rejected in production. (KPI card)

- Trend of Reject Rates: Monthly/quarterly trends of reject rates to detect patterns and identify process-related issues. (line chart)
- Root Cause Analysis: Breakdown of reject rates by different categories (e.g., product type, production line, equipment). (Stacked Column Chart)
- Reject Rate by Site/Line (*Clustered Bar Chart*)
- Equipment Uptime: Compare uptime % across machines or lines. (Donut Chart)
- Maintenance Frequency: Frequency and costs associated with maintenance or breakdowns for machines. (Bar Chart)
- Visualization of line changeover time: Time lost during product switchovers per site. (Line Chart over time)

Additional metrics are reported as per the different requirements for respective tools developed, ex. SUD bag reject tool (focuses on bag rejects causes), SUD leakage tool (focuses on leakage in pods and its overall impact), etc.

Technologies Used: Python, Pandas, SQL, Power BI

Impact:

The project contributed to improvements in manufacturing operations at P&G by uncovering key insights that helped to reduce equipment failures and reject rates, resulting in cost savings and enhanced operational efficiency.